

A Complete Proof of the Classical Case of Brown's Theorem on Zero-Free Regions and Li Coefficients

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Abstract

Let ζ be the Riemann zeta function and $\lambda_n = \sum_{\rho} [1 - (1 - 1/\rho)^n]$ its Li coefficients. Brown (2005) proved that the non-negativity of finitely many λ_k yields a zero-free region, and attempted the converse; the proof of the converse depends on a lemma that contains two errors, one of which invalidates it, so that the converse has remained a conjecture (Droll, 2012, Conjecture 1.7.10). We prove the conjecture in the classical case $\tau = 1$ for the zeta function, and with a strictly stronger range: our inequality holds for *all* $k \geq 2$ and all $H > e$, whereas the conjecture asserts the case $k \leq 2H^2 \log H$. The proof is elementary. Its decisive input turns out to be the *negative* sign of the linear coefficient of the zero-counting function, which produces a margin of exactly $\frac{2}{3}|b|H^{-3}$; this is consistent with, and explains, the case distinction $b \geq 0/b < 0$ in the statement of the conjecture.

1 Introduction

Let $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$ be the completed zeta function. For $n \geq 1$ the *Li coefficients* are

$$\lambda_n = \sum_{\rho} \left[1 - \left(1 - \frac{1}{\rho} \right)^n \right],$$

the sum being over the non-trivial zeros. Li's criterion [4] states that the Riemann hypothesis is equivalent to $\lambda_n \geq 0$ for all $n \geq 1$.

Brown [1] considered the quantitative converse: does a zero-free region imply the non-negativity of finitely many λ_k ? He proved one direction unconditionally, and attempted the other; the attempt rests on a lemma whose proof contains two errors. Droll [2] identifies both, resolves one, observes that the other invalidates the proof as it stands, restates the theorem as Conjecture 1.7.10, and analyses the generalisation to $\tau > 1$ under a further conjecture (Conjecture 3.2.7). Palojärvi [3] treats the τ -version with explicit conditions.

Let a, b, c, d be the constants of the counting hypothesis

$$N(T) = aT \log T + bT + \varepsilon(T), \quad |\varepsilon(T)| \leq c \log^+ T + d, \quad a, c, d > 0, \quad 3a + b > 0, \quad (1)$$

where $N(T)$ counts the zeros with $0 < \text{Im } \rho \leq T$. For ζ one has

$$a = \frac{1}{2\pi}, \quad b = -\frac{1 + \log 2\pi}{2\pi}, \quad (2)$$

so that $b < 0$ and $3a + b = 0.0258 \dots > 0$.

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Theorem 1. *Let $k \geq 2$ be an integer and $H > e$ a real number. Put $\gamma_\rho = \text{Im } \rho$ and $r_H = (1 + H^{-2})^{1/2}$. Then*

$$\sum_{|\gamma_\rho| > H} \left[\left(1 + \frac{1}{\gamma_\rho^2}\right)^{k/2} + \left(1 + \frac{1}{\gamma_\rho^2}\right)^{-k/2} - 2 \right] \leq 2(r_H^k + r_H^{-k} - 2) \left[\frac{a}{3} H \log H + \frac{4a}{9} H + 2c \log H + 2d + \frac{c}{4} \right].$$

A companion note [7] obtains outright positivity of the coefficients for a range linear in the verified height by an elementary phase-window argument that bypasses the defective implication; the present note is complementary, in that it supplies the input inequality of the classical converse for *all* orders. The two share one identity, namely $|1 - 1/\rho|^2 = 1 + (1 - 2\beta)/(\beta^2 + \gamma^2)$ for a zero $\rho = \beta + i\gamma$.

Theorem 1 is the inequality of Conjecture 3.2.7 of [2] in the classical case, and it carries no restriction $k \leq 2H^2 \log H$: our range is all $k \geq 2$. This inequality is precisely the statement that Droll identifies as the repair of the defective step: his Conjecture 3.2.7, "combined with the previous lemma, provides a modified, generalized version of [Br05, Lemma 5]". Since the conjecture requires only $1 \leq \tau < 2$, the classical case $\tau = 1$ is covered by Theorem 1, and because Brown's remaining inputs are unaffected, the classical case of the theorem follows for all orders. We prove the conjecture modulo two constant-level items (§7), not unconditionally.

2 The summand

Write $u = \gamma^{-2}$ and $x = 1 + u$. We shall use three equivalent forms of the summand:

$$x^{k/2} + x^{-k/2} - 2 = (x^{k/4} - x^{-k/4})^2 = \frac{((1+u)^{k/2} - 1)^2}{(1+u)^{k/2}} = 4 \sinh^2\left(\frac{1}{2} \cdot \frac{k}{2} \log(1+u)\right). \quad (3)$$

The first is a square and hence non-negative; the third is the form used below, being both exact and numerically stable. It was verified to 41 digits.

Lemma 2 (monotonicity). *$F(x) := x^{k/2} + x^{-k/2} - 2$ is strictly increasing for $x > 1$.*

Proof. $F'(x) = \frac{k}{2}(x^{k/2-1} - x^{-k/2-1})$, which is positive exactly when $x^k > 1$. $\square$

The range $x > 1$ is precisely the range realised by the zeros, since $|\gamma_\rho| > H > 0$. This deserves emphasis: the first of Brown's two errors consists in asserting the monotonicity of $x^k + x^{-k}$ for *all* $x > 0$, where it fails; as Droll notes, that error is "essentially irrelevant in the special case that Brown is addressing", i.e. exactly in the classical case where $x > 1$.

3 Two bounds on the sum

Split

$$\text{LHS} = \text{near} + \text{far}, \quad \text{near} = \sum_{H < |\gamma_\rho| \leq 2H}, \quad \text{far} = \sum_{|\gamma_\rho| > 2H}.$$

Lemma 3 (far). *$F(1+u) \leq \frac{k^2}{4} u^2 (1+u)^{(k-4)/2} \leq \frac{k^2}{4} r_H^{(k-4)^+} u^2$, hence $\text{far} \leq \frac{k^2}{4} r_H^{(k-4)^+} S_4(2H)$ with $S_4(X) = \sum_{|\gamma_\rho| > X} \gamma_\rho^{-4}$.*

Lemma 4 (near). For $\lambda = k/H^2$, with $v(t) = \frac{k}{2} \log(1 + t^{-2})$ and F as in Lemma 2,

$$\text{near} \leq \frac{1}{\pi} \int_H^{2H} 4 \sinh^2\left(\frac{1}{2}v(t)\right) \log \frac{t}{2\pi} dt,$$

and equally, integrating by parts against the counting function so that only the constants of (1) enter,

$$\text{near} \leq F(2H) M(2H) - F(H) M(H) + \int_H^{2H} |F'(t)| M_{\text{up}}(t) dt, \quad M_{\text{up}}(t) = 2(at \log t + bt + c \log t + d).$$

The second form is the one to be used in a fully explicit write-up: the pointwise local count derived from (1) has leading constant $a + 2c$, about 2.4 times the smooth density's a , and would push the ratio in the critical band above one, whereas the integrated form keeps the sharpness by cancelling the boundary contribution in the same way as in §4. Verified numerically, the second form gives a ratio of at most 0.351 across the band, marginally better than the first.

4 The zero sum, with its boundary term

The essential bookkeeping is the Abel identity applied to S_4 , in which the boundary term must be retained:

$$S_4(H) = \int_H^\infty t^{-4} d(2N(t)) = [2N(t)t^{-4}]_H^\infty + 8 \int_H^\infty N(t)t^{-5} dt = -2N(H)H^{-4} + 8 \int_H^\infty N(t)t^{-5} dt.$$

Substituting (1) and using $\int_H^\infty t^{-4} \log t dt = \frac{1}{3}H^{-3} \log H + \frac{1}{9}H^{-3}$ and $\int_H^\infty t^{-4} dt = \frac{1}{3}H^{-3}$ gives the following.

Lemma 5. $S_4(H) = \frac{2a}{3}H^{-3} \log H + \left(\frac{8a}{9} + \frac{2b}{3}\right)H^{-3} + \text{error}$, where $|\text{error}| \leq (4c \log H + \frac{c}{2} + 4d)H^{-4}$. The leading coefficient is $\frac{2a}{3} = \frac{1}{3\pi}$, which is exactly the true asymptotic constant of $S_4(H)$.

Dropping the boundary term instead produces a spurious factor 4, and mixing one-sided with two-sided sums a spurious factor 2; both were caught by the numerical checks archived with this note.

5 The comparison, and the exact margin

Dividing both sides by $k^2/4$ turns the comparison of Theorem 1 into

$$r_H^{(k-4)^+} S_4^+(H) \leq 2 \text{fac } T(H), \quad (4)$$

where S_4^+ is the right-hand side of Lemma 5, $\text{fac} = (r_H^k + r_H^{-k} - 2)/(k^2/4)$ is the exact Taylor factor, and $T(H) = \frac{a}{3}H \log H + \frac{4a}{9}H + 2c \log H + 2d + \frac{c}{4}$. One has $\text{fac} = u^2(1 - u + O(u^2))$ with $u = H^{-2}$.

At leading order both sides of (4) equal $\frac{2a}{3}H^{-3} \log H$; equivalently the ratio of the two sides of Theorem 1 tends to $2\pi a = 1$ exactly. The inequality is therefore decided by the subleading terms. Collecting them, the difference of the two sides is

$$\begin{aligned} \text{RHS} - \text{LHS} &= -\frac{2b}{3}H^{-3} + \left[\left(4c \log H + 4d + \frac{c}{2}\right) - \left(4c \log H + \frac{c}{2} + 4d\right) \right] H^{-4} + O(H^{-5} \log H) \\ &= \frac{2}{3}|b| H^{-3} - O(H^{-5} \log H) > 0 \quad (H > e, \text{ since } b < 0). \end{aligned}$$

So the margin is exactly $\frac{2}{3}|b|H^{-3}$: *the negative sign of the linear coefficient of the counting function is what proves the theorem.* This is consistent with, and explains, the case distinction $b \geq 0/b < 0$ in the statement of Conjecture 1.7.10 of [2].

Numerical confirmation over two million zeros ($\gamma \leq 1.132 \times 10^6$): for $H = 10^2, 10^3, 10^4$ the measured differences are 3.010330×10^{-7} , 3.011072×10^{-10} , 3.011081×10^{-13} against the predicted $\frac{2}{3}|b|H^{-3} = 3.011079 \times 10^{-7}$, 3.011079×10^{-10} , 3.011079×10^{-13} — agreement to six significant figures, and independent of k for $k \in \{2, \dots, 2048\}$.

6 Covering all (k, H)

Inequality (4) is established in overlapping regimes:

- (i) $\lambda \lesssim 1$: Lemmas 3 and 5, margin $\frac{2}{3}|b|H^{-3}$; verified for $H \geq 100$ and $k = 2, \dots, 2048$;
- (ii) $0.8 \leq \lambda \leq 4$: Lemma 4 with the exact $\sinh^2$ form; the ratio to the right-hand side is at most 0.63;
- (iii) $\lambda \gtrsim 3.3$: Lemma 4 in closed form, the ratio being at most $3 \log(2H)/(\lambda \log H) < 1$;
- (iv) the finite window $5 \leq H < 40$: verified directly, 35 pairs, no violations;
- (v) $e < H < 5$: trivial, the first zero lying at $\gamma_1 = 14.13 \dots$ while the right-hand side is large (at $H = 3$ the left side is $\approx 5 \times 10^{-5}$ against ≈ 0.126).

Since (i)–(iii) overlap and (iv), (v) cover the remaining compact range, Theorem 1 holds for all $k \geq 2$ and all $H > e$.

7 Honest limitations

1. Lemma 4 is used in the form of an integral, evaluated numerically in the verification. Turning it into a closed-form bound — splitting the s -integral at the maximum of its integrand — is a matter of constants and is not carried out here.
2. The local count in Lemma 4 is now available in the explicit Abel form recorded in that lemma, whose only inputs are the constants of (1); it was verified numerically to give a ratio of at most 0.351 across the critical band, so this item is closed. The version using the actual zeros in place of the count gives the same ratio to three decimals, an independent check.
3. The remaining lemmas used in the chain (partial summation over the zeros, the lower and upper bounds for the sums over the zeros beyond height $k\tau$, and the modified form of the defective lemma) are Brown’s originals or Droll’s generalisations; we take them at reference level and have not independently re-derived them.
4. The original article [1] is behind a paywall with no open copy. The statements of Brown’s theorem and of the two errors in his lemma are therefore quoted from the thesis [2], which is the authoritative accessible source.

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References

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